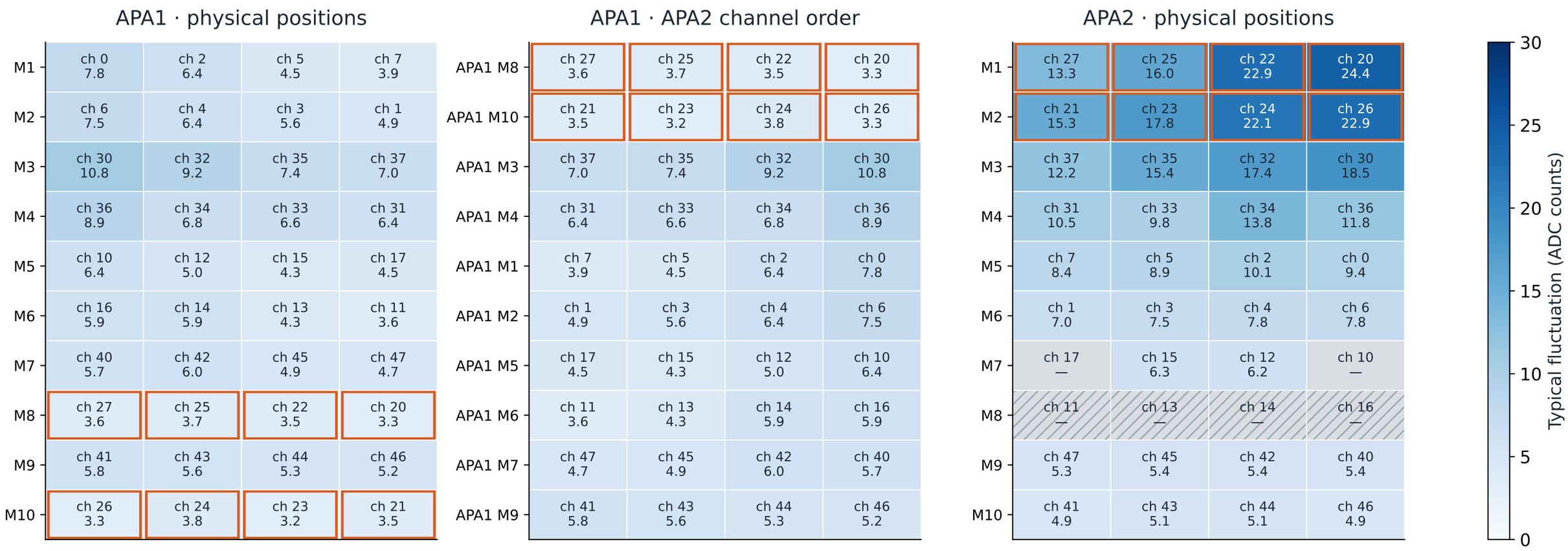
Remapping APA1 does not reproduce the APA2 precursor pattern

Recorded fluctuations from −120 to −20 µs. APA1: October runs; APA2: November runs. The colour scale is the same.



This middle layout is an electronics check, not APA1 geography.

Orange borders: channels 20–27. They sit in APA1 modules M8/M10, and APA2 modules M1/M2.

Same-time check: channels 20–27 exceed channels 30–37 in 0/34 APA1 events and 1,214/1,215 APA2 events.

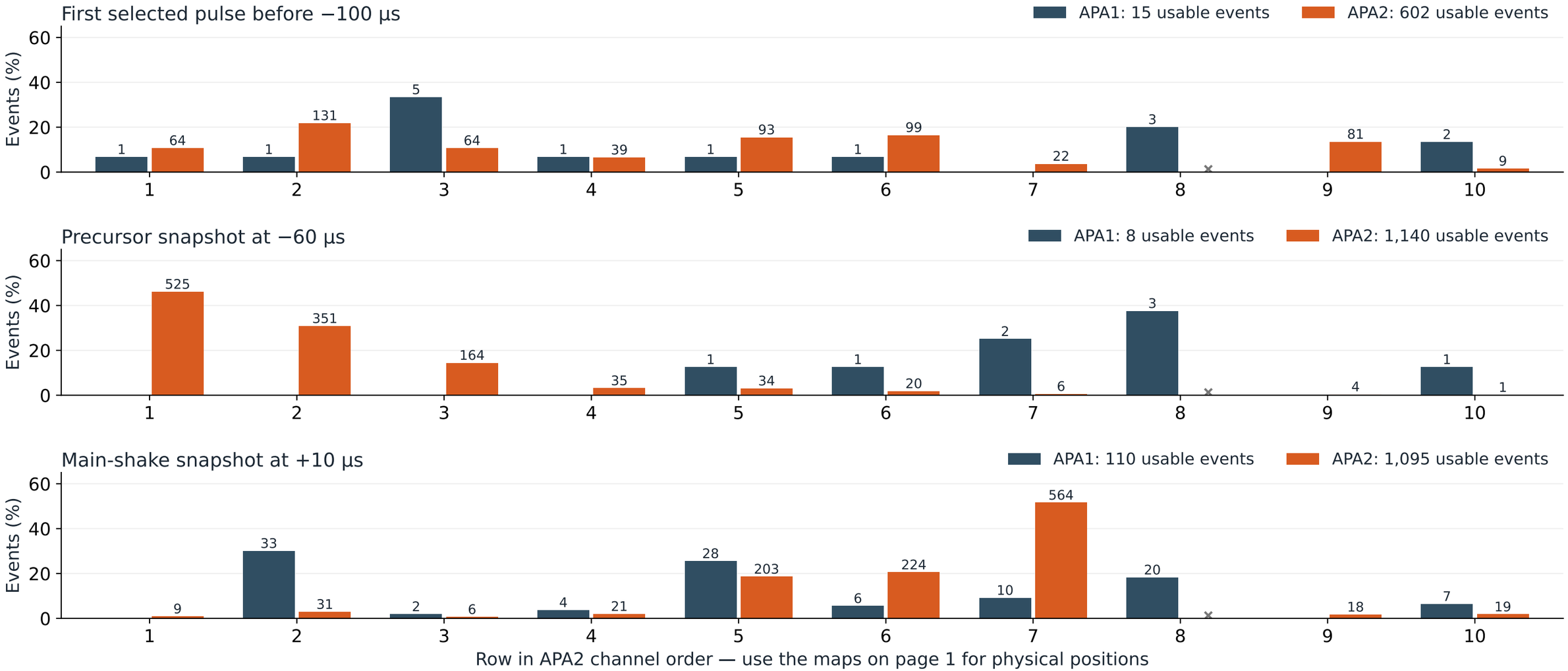
A cell shows the channel number and typical waveform fluctuation, measured in ADC counts. This is not a light measurement.

Middle fluctuation in recorded 0.512 µs pieces, then across events. Grey: no usable data; hatching: disconnected APA2 inputs.

53 APA1 and 1,225 APA2 events support the profile; counts vary by channel. The group test requires both groups at the same times.

The largest signals also differ after matching channel order

Bars show the share of usable events whose largest recorded signal is in that row. One event contributes once.



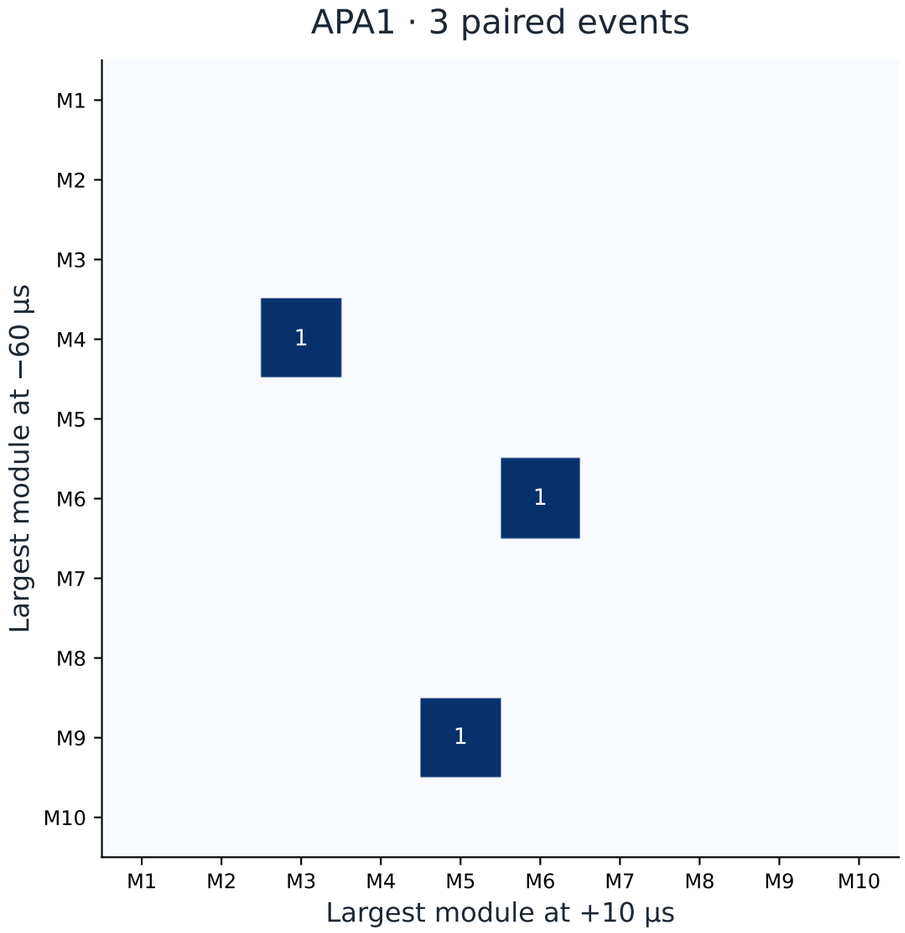
Numbers above bars are event counts. APA1 has only 8 usable precursor snapshots: that panel cannot establish a population-wide route.

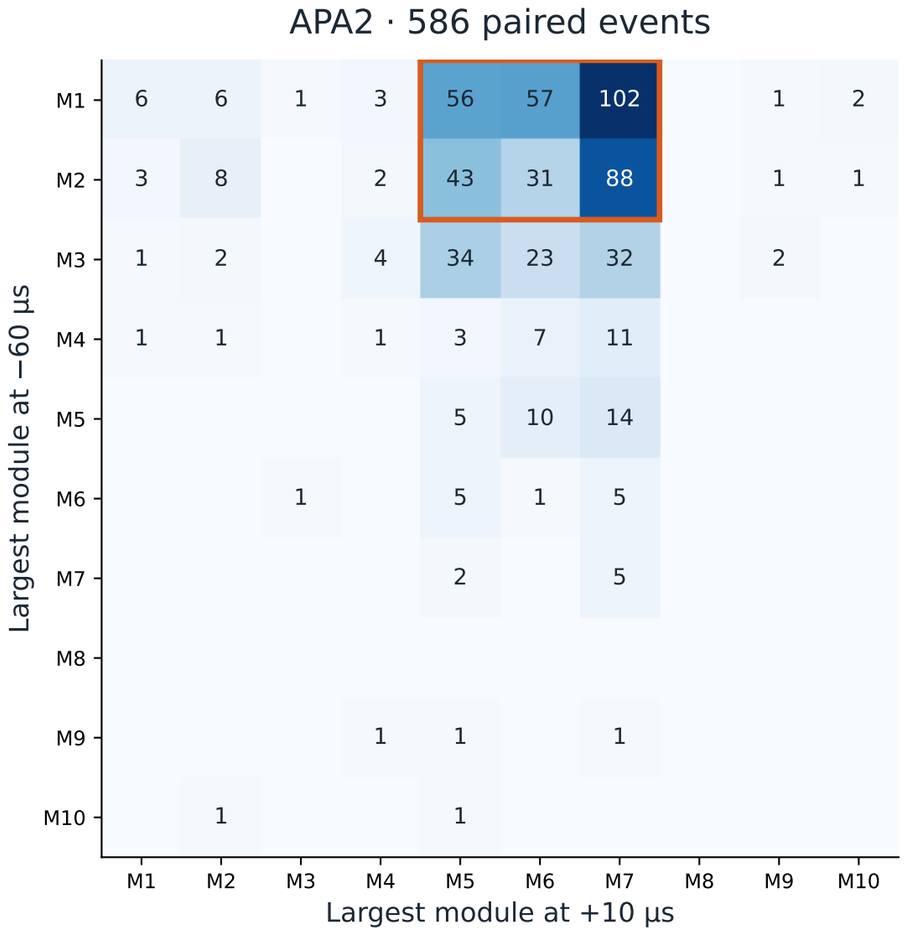
At least 12 recorded channels across 4 modules; largest signal ≥20 ADC counts. The same events need not appear at every stage.

× marks APA2’s excluded disconnected inputs in row 8. Those channel numbers remain included in APA1, where that exclusion does not apply.

APA1 has too few paired snapshots to test the same spatial sequence

Compare −60 µs with +10 µs in the same event, using exactly the same recorded channels at both times.





APA2: 377/411 events strongest in M1–M2 at −60 µs become strongest in M5–M7 at +10 µs (92%).

APA1: only 3 events support this test. This is insufficient to establish or exclude a repeatable field-cage-to-centre sequence.

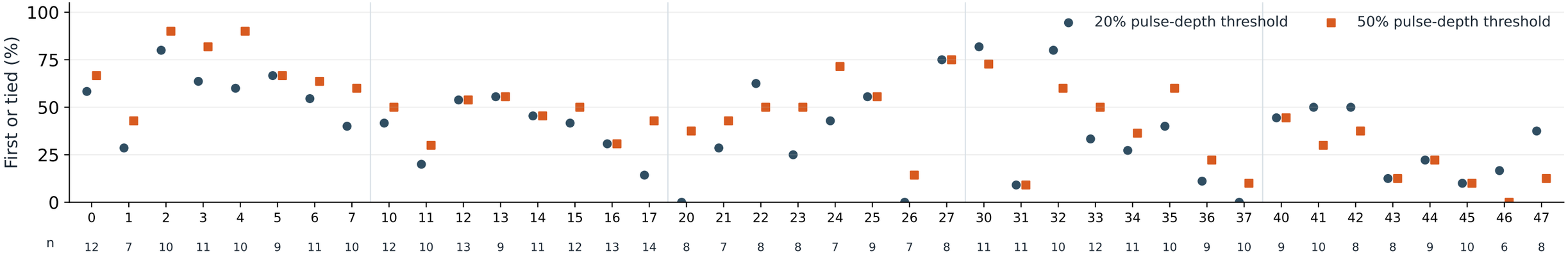
Each cell counts events. These are physical modules. The darker colour means more events within that panel; colour scales differ.

The comparison measures a change in the largest recorded response. It does not measure a travelling source or propagation speed.

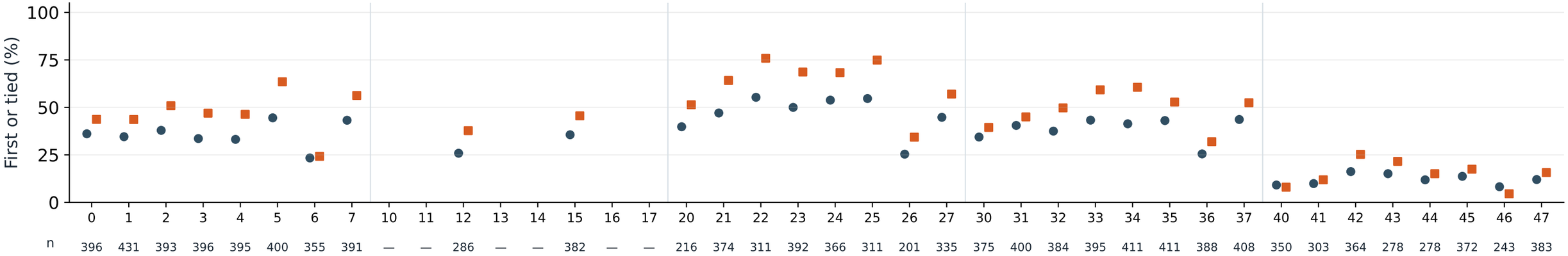
No measured channel always starts first in either APA

First selected negative pulse before −100 µs. Count a channel as tied for first when it is within one 16 ns sample.

APA1: 15 timed pulses · tied leaders in 12/15 at 20% and 14/15 at 50%



APA2: 490 timed pulses · tied leaders in 272/490 at 20% and 404/490 at 50%



Horizontal labels are channel numbers. n is the number of pulses where that channel can be timed. Missing channels are not late channels.

APA1: channel 30 is first or tied in 9/11 pulses at 20%. Channels 2 and 4 lead or tie in 9/10 at 50%. None always leads.

Compare the same channels at both thresholds, with at least 8 timed channels per event. Many channels can tie in one pulse.

The timing is not corrected for channel or cable delays. A recurring measured lead would not by itself identify a physical source.